

Lab06__Barbosa

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Laboratório 3: SQLite

1. Baixe o arquivo `disco.db` e armazene na variável `path` o caminho completo (pasta) na qual o arquivo foi gravado. Utilize o comando `file.path()` para combinar a variável `path` com o nome do arquivo (`disco.db`) e obter o nome do arquivo com seu respectivo caminho. Armazene este resultado na variável `fname`.

```
library(RSQLite)

path <- "."
fname <- file.path(path, "disco.db")
```

2. Utilizando o pacote `RSQLite`, conecte-se ao arquivo de banco de dados. Armazene a conexão na variável `conn`.

```
conn <- dbConnect(SQLite(), fname)
```

3. Liste as tabelas existentes no banco de dados.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
tbl(conn, "sqlite_master") %>%
  filter(type == "table") %>%
  arrange(name) %>%
  pull(name)
```

```
[1] "albums"          "artists"          "customers"         "employees"
[5] "genres"          "invoice_items"    "invoices"          "media_types"
[9] "playlist_track"  "playlists"        "sqlite_sequence"   "sqlite_stat1"
[13] "tracks"
```

4. Identifique os nomes de todas as colunas existentes na tabela `customers`.

```
tbl(conn, "customers") %>%
  colnames()
```

```
[1] "CustomerId"  "FirstName"  "LastName"  "Company"  "Address"
[6] "City"       "State"     "Country"   "PostalCode" "Phone"
[11] "Fax"        "Email"     "SupportRepId"
```

5. Utilizando **apenas** SQLite, com o apoio do comando `dbGetQuery`, identifique quantos clientes estão atualmente cadastrados neste banco de dados.

```
dbGetQuery(conn, "SELECT COUNT(*) as total_clientes FROM customers")
```

```
total_clientes
1              59
```

6. Utilizando **apenas** SQLite, identifique o número de países diferentes em que moram os clientes encontrados acima.

```
dbGetQuery(conn, "SELECT COUNT(DISTINCT Country) as countries
                  FROM customers")
```

```
countries
1         24
```

7. Utilizando **apenas** SQLite, quantos clientes existem por país? A tabela resultante deve conter o nome do país e a respectiva contagem, além de ser ordenada de maneira decrescente pela referida contagem.

```
dbGetQuery(conn, "SELECT Country, COUNT(*) as Quantity FROM customers
                  GROUP BY Country
                  ORDER BY Quantity DESC")
```

	Country	Quantity
1	USA	13
2	Canada	8
3	France	5
4	Brazil	5
5	Germany	4
6	United Kingdom	3
7	Portugal	2
8	India	2
9	Czech Republic	2
10	Sweden	1
11	Spain	1
12	Poland	1
13	Norway	1
14	Netherlands	1
15	Italy	1
16	Ireland	1
17	Hungary	1
18	Finland	1
19	Denmark	1
20	Chile	1
21	Belgium	1
22	Austria	1
23	Australia	1
24	Argentina	1

8. Quais são os 5 países com mais clientes registrados? Use apenas SQLite.

```
dbGetQuery(conn, "SELECT Country, COUNT(*) as Quantity FROM customers
                  GROUP BY Country
                  ORDER BY Quantity DESC
                  LIMIT 5")
```

	Country	Quantity
1	USA	13
2	Canada	8
3	France	5

4	Brazil	5
5	Germany	4

9. Quais são os países registrados que possuem apenas 6 letras no nome?

```
dbGetQuery(conn, "SELECT DISTINCT Country as Countries FROM customers
                  WHERE LENGTH(Country) = 6")
```

	Countries
1	Brazil
2	Canada
3	Norway
4	France
5	Poland
6	Sweden

10. Quais foram as músicas compradas por clientes brasileiros?

```
musicas_brasil <- tbl(conn, "customers") %>%
  filter(Country == "Brazil") %>%
  inner_join(tbl(conn, "invoices"), by = "CustomerId") %>%
  inner_join(tbl(conn, "invoice_items"), by = "InvoiceId") %>%
  inner_join(tbl(conn, "tracks"), by = "TrackId") %>%
  select(TrackName = Name) %>%
  distinct() %>%
  arrange(TrackName) %>%
  pull(TrackName)

musicas_brasil
```

```
[1] "1/2 Full"
[2] "2 Minutes To Midnight"
[3] "A Cor Do Sol"
[4] "A Cura"
[5] "A Menina Dança"
[6] "Abraham, Martin And John"
[7] "Aces High"
[8] "Admirável Gado Novo"
[9] "All Along The Watchtower"
[10] "All I Want Is You"
[11] "Aloha"
[12] "Amor De Muito"
```

- [13] "Animal"
- [14] "Any Colour You Like"
- [15] "Aos Leões"
- [16] "Babyface"
- [17] "Back off Bitch"
- [18] "Bad"
- [19] "Bad Seed"
- [20] "Banditismo Por Uma Questa"
- [21] "Be Good Johnny"
- [22] "Be Mine"
- [23] "Big Wave"
- [24] "Binky The Doormat"
- [25] "Bittersweet Me"
- [26] "Black"
- [27] "Black Light Syndrome"
- [28] "Bora-Bora"
- [29] "Bossa"
- [30] "Burden In My Hand"
- [31] "Calling Dr. Love"
- [32] "Carolina Hard-Core Ecstasy"
- [33] "Casa"
- [34] "Caso Você Queira Saber"
- [35] "Child In Time"
- [36] "Cinema Mudo"
- [37] "Cold Gin"
- [38] "Coma"
- [39] "Communication Breakdown(2)"
- [40] "Cristina Nº 2"
- [41] "Cropduster"
- [42] "Cry For Love"
- [43] "Cérebro Eletrônico"
- [44] "D'Yer Mak'er"
- [45] "Damage Inc."
- [46] "Dance"
- [47] "Demorou!"
- [48] "Dissident"
- [49] "Don't Cry (Original)"
- [50] "Don't Look Back"
- [51] "Doutor"
- [52] "Down Under"
- [53] "Down by the Sea"
- [54] "Drifter"
- [55] "Duelists"

- [56] "Dust N' Bones"
- [57] "Electrolite"
- [58] "Eu Amo Você"
- [59] "Everything I Need"
- [60] "Experiment In Terra"
- [61] "Eye"
- [62] "Fantasia On Greensleeves"
- [63] "Fast As a Shark"
- [64] "Flower"
- [65] "Flying High Again"
- [66] "Fool In The Rain"
- [67] "Garden of Eden"
- [68] "Get Down, Make Love"
- [69] "Get Up"
- [70] "Girl From A Pawnshop"
- [71] "Go Down"
- [72] "Green Disease"
- [73] "Grito De Alerta"
- [74] "Heart Of Lothian: Wide Boy / Curtain Call"
- [75] "Hell Ain't A Bad Place To Be"
- [76] "Help Yourself"
- [77] "Home Sweet Home"
- [78] "I Belong To You"
- [79] "I Feel Good (I Got You) - Sossego"
- [80] "In Bloom"
- [81] "In The Evening"
- [82] "Interlude Zumbi"
- [83] "Into The Light"
- [84] "Intro"
- [85] "Ipiranga 2001"
- [86] "Is This Love (Live)"
- [87] "It's a Mistake"
- [88] "Jesus Christ Pose"
- [89] "Karelia Suite, Op.11: 2. Ballade (Tempo Di Menuetto)"
- [90] "Knocking At Your Back Door"
- [91] "Lay Down Sally"
- [92] "Leash"
- [93] "Leper Messiah"
- [94] "Like A Song..."
- [95] "Lindo Lago Do Amor"
- [96] "Linha Do Equador"
- [97] "Live and Let Die"
- [98] "Lixo Do Mangue"

- [99] "Losfer Words"
- [100] "Mama, I'm Coming Home"
- [101] "Maracatu De Tiro Certeiro"
- [102] "Mateus Enter"
- [103] "Maybe I'm A Leo"
- [104] "Mensagen De Amor (2000)"
- [105] "Meu Caro Amigo"
- [106] "Meu Erro"
- [107] "Midnight From The Inside Out"
- [108] "Mis Penas Lloraba Yo (Ao Vivo) Soy Gitano (Tangos)"
- [109] "Morena De Angola"
- [110] "Most High"
- [111] "Mundaréu"
- [112] "Music for the Funeral of Queen Mary: VI. \"Thou Knowest, Lord, the Secrets of Our Heart"
- [113] "Nega Do Cabelo Duro"
- [114] "No No No"
- [115] "No Quarter"
- [116] "Nossa Gente (Avisa Lá)"
- [117] "Nosso Adeus"
- [118] "Não Vou Ficar"
- [119] "O Descobridor Dos Sete Mares"
- [120] "O Leaozinho"
- [121] "O Que Me Importa"
- [122] "Oceans"
- [123] "On A Plain"
- [124] "Pais E Filhos"
- [125] "Paranoid"
- [126] "Pick Myself Up"
- [127] "Plot 180"
- [128] "Podes Crer"
- [129] "Ponto De Interrogação"
- [130] "Powerslave"
- [131] "Praise"
- [132] "Red Light"
- [133] "Redundant"
- [134] "Refavela (Live)"
- [135] "Reggae Tchan"
- [136] "Revolta Olodum"
- [137] "Right Next Door to Hell"
- [138] "Rios Pontes & Overdrives"
- [139] "Rock And Roll Is Dead"
- [140] "Rocket Queen"
- [141] "Saber Amar"

[142] "Samba Do Lado"
 [143] "Sanctuary"
 [144] "Saudade Dos Aviões Da Panair (Conversando No Bar)"
 [145] "Scentless Apprentice"
 [146] "Selvagem"
 [147] "Sereia"
 [148] "Será"
 [149] "Será Que Vai Chover?"
 [150] "Seven Seas Of Rhye"
 [151] "Shakes and Ladders"
 [152] "Shout It Out Loud"
 [153] "Sincero Breu"
 [154] "Slither"
 [155] "Snowballed"
 [156] "So Fast, So Numb"
 [157] "Sometimes Salvation"
 [158] "Soul Singing"
 [159] "Stand Inside Your Love"
 [160] "Stir It Up (Live)"
 [161] "String Quartet No. 12 in C Minor, D. 703 \"Quartettsatz\": II. Andante - Allegro assa.
 [162] "Strutter"
 [163] "Suite No. 3 in D, BWV 1068: III. Gavotte I & II"
 [164] "Sweetest Thing"
 [165] "Symphony No. 2, Op. 16 - \"The Four Temperaments\": II. Allegro Comodo e Flemmatico"
 [166] "Take the Celestra"
 [167] "The Day I Tried To Live"
 [168] "The Great Gig In The Sky"
 [169] "The Memory Remains"
 [170] "The Unforgiven II"
 [171] "The Unwritten Law"
 [172] "Think About You"
 [173] "Title Song"
 [174] "Top Top"
 [175] "TriboTchan"
 [176] "Um Satélite Na Cabeça"
 [177] "Untitled"
 [178] "Vai Passar"
 [179] "Vai Valer"
 [180] "Vamo Batê Lata"
 [181] "Vavoom : Ted The Mechanic"
 [182] "Voce Nao Entende Nada - Cotidiano"
 [183] "Wanted Dread And Alive"
 [184] "Why Go"

```
[185] "X-9 2001"  
[186] "You're My Best Friend"  
[187] "You've Been A Long Time Coming"  
[188] "Your Blue Room"  
[189] "Água de Beber"
```

11. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-16 00:53:29 -03"
```